

3. Evaluate the following limits (KU: 8 marks)

$$\lim_{x \rightarrow -4} \frac{\sqrt{x^2 + 9} - 5}{x + 4} f(x)$$

0 so must be conjugate

$$\lim_{x \rightarrow -4} \frac{\sqrt{x^2 + 9} - 5}{(x+4)(\sqrt{x^2+9} + 5)}$$

$$\lim_{x \rightarrow -4} \frac{(x+4)(x-4)}{(x+4)(\sqrt{x^2+9} + 5)}$$

$$\lim_{x \rightarrow -4} \frac{(x-4)}{\sqrt{x^2+9} + 5} = \frac{-4}{10} = -\frac{2}{5}$$

$$\lim_{x \rightarrow -4} f(x) = -\frac{2}{5}$$

$$\lim_{x \rightarrow -\infty} \left[\frac{(2x^2 + 1)^2}{2x(x^2 + x)^2} \right]$$

Find end behaviour model

$$\frac{4x^4}{2x^5}$$

approaches 0 since

numerator < denominator

Plugging in large value

for $x = 100$ results in

a slightly negative

value

4. Using the definitions of the horizontal and vertical asymptotes, determine HA and VA of the graph of:

$$f(x) = \frac{\sqrt{2x^2 + 1}}{3x - 5}$$

VA is when

$$\lim_{x \rightarrow c^-} f(x) = \pm \infty$$

$$\lim_{x \rightarrow c^+} f(x) = \pm \infty$$

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point of interest is $\frac{5}{3}$

$$\lim_{x \rightarrow \frac{5}{3}^+} \frac{\sqrt{2x^2 + 1}}{3x - 5} = \infty$$

$$\lim_{x \rightarrow \frac{5}{3}^-} \frac{\sqrt{2x^2 + 1}}{3x - 5} = -\infty$$

$$\lim_{x \rightarrow \frac{5}{3}^-} \frac{\sqrt{2x^2 + 1}}{3x - 5} = -\infty$$

Thus there is a VA

at $x = \frac{5}{3}$

(KU: 3 marks)

1.5

A horizontal asymptote exists when $\lim_{x \rightarrow \infty} f(x) = a$ or $\lim_{x \rightarrow -\infty} f(x) = a$ HA

$$\lim_{x \rightarrow \infty} f(x) = \frac{\sqrt{2x^2 + 1}}{3x - 5}$$

End behaviour model is

$$\lim_{x \rightarrow \infty} f(x) = \frac{\sqrt{2} |x|}{3x} = \frac{\sqrt{2}}{3}$$

$$\lim_{x \rightarrow -\infty} f(x) = \frac{\sqrt{2} |x|}{3x} = -\frac{\sqrt{2}}{3}$$

$$\lim_{x \rightarrow -\infty} f(x) = -\frac{\sqrt{2}}{3}$$

$$\lim_{x \rightarrow 1} \frac{2x^3 - 3x^2 + 1}{x^3 - x^2 - x + 1}$$

Factor top and bottom

$$\lim_{x \rightarrow 1} \frac{(x-1)(x-1)(2x+1)}{x^2(x-1) - 1(x-1)}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(x-1)(2x+1)}{(x+1)(x-1)(x-1)}$$

$$\lim_{x \rightarrow 1} \frac{2x+1}{x+1} = \text{substitute}$$

$$= \frac{3}{2}$$

$$\lim_{x \rightarrow 2} f(x) \text{ where } f(x) = \begin{cases} 2x-1 & \text{if } x < 2 \\ -3 & \text{if } x = 2 \\ 1-x^2 & \text{if } x > 2 \end{cases}$$

$\lim_{x \rightarrow 2} f(x)$ exists if $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (2x-1)$$

$$= \lim_{x \rightarrow 2^-} (1-x^2)$$

$$= -3$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} (1-x^2)$$

$$= -3$$

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5. Determine the y-intercept of the tangent at $x = 5$ of the function $f(x) = \frac{1}{x^2 + x - 2}$. (A: 3 marks)

$$M_T(s) = \lim_{a \rightarrow x} \frac{f(x) - f(a)}{x - a}$$

at $x = 5$

$$f(5) = \frac{1}{25 + 5 - 2} = \frac{1}{28}$$

$$= \lim_{a \rightarrow 5} \frac{\frac{1}{(x+2)(x-1)} - \frac{1}{(a+2)(a-1)}}{x - a}$$

$$= \lim_{a \rightarrow 5} \left(\frac{a^2 + a + x^2 - x}{(x+2)(x-1)(a+2)(a-1)} \right)$$

$$= \lim_{a \rightarrow 5} \frac{(a-x)(x-a)}{(x+2)(x-1)(a+2)(a-1)}$$

$$= \frac{-(x-a)}{(x+2)(x-1)(a+2)(a-1)}$$

$$= \frac{1}{(x+2)(x-1)(a+2)(a-1)}$$

$$= \frac{-(x+a+1)}{(x+2)(x-1)(a+2)(a-1)} = M_T(s)$$

$$= \frac{-(5+5+1)}{(5+2)(5-1)(5+2)(5-1)}$$

$$= \frac{-11}{784} = M_T(s)$$

$$y = \frac{11}{784}(x-5) + \frac{1}{28}$$

(or)

$$y = \frac{11}{784}x - \frac{27}{784}$$

$$y = \frac{11}{784}x - \frac{27}{784}$$

6. The displacement in metres of a particle moving in a straight line is given by $s(t) = t^2 - 8t + 18$, where t is measured in seconds. Determine the instantaneous velocity when $t = 4$ s.

Instantaneous velocity at $t = 4$

$$= \lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h} \quad t = 4$$

$$= \lim_{h \rightarrow 0} \frac{(t+h)^2 - 8(t+h) + 18 - (t^2 - 8t + 18)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{t^2 + 2th + h^2 - 8t - 8h + 18 - t^2 + 8t - 18}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2th + h^2 - 8h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(2t + h - 8)}{h}$$

$$= \lim_{h \rightarrow 0} (2t + h - 8)$$

$$= 2t - 8 \quad t = 4$$

Instantaneous velocity is 0 m/s

Use the intermediate value theorem to show that there is a root of the given equation

$$y = x^4 + x - 3 \text{ in interval } (1, 2)$$

If $f(x)$ is continuous, then there exists a value c such that

$$f(1) < f(c) < f(2)$$

$$f(1) = -1$$

$$f(2) = 15$$

$$f(1) < f(c) < f(2)$$

Since 0 lies between $f(1)$ and $f(2)$

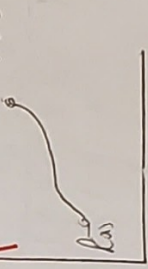
there is at least one c such that $f(c) = 0$

in the range $c \in [1, 2]$

(-1)

(C: 2 marks)

1.5



8. Using the squeeze theorem prove that: $\lim_{x \rightarrow 0^+} \sqrt{x} e^{\sin(\frac{\pi}{x})} = 0$ (KU: 3 marks)

(Hint: $y = e^x$ is an increasing function)

$e^{\sin(\frac{\pi}{x})}$ is bounded between e^{-1} and e^1 because $\sin(\frac{\pi}{x})$ is

$$e^{-1} \leq e^{\sin(\frac{\pi}{x})} \leq e^1$$

$$\sqrt{x} e^{-1} \leq \sqrt{x} e^{\sin(\frac{\pi}{x})} \leq \sqrt{x} e^1$$

Now we can take the limit as $x \rightarrow 0$

$$0 \leq \sqrt{x} e^{\sin(\frac{\pi}{x})} \leq 0$$

$$\lim_{x \rightarrow 0} \sqrt{x} e^{\sin(\frac{\pi}{x})} = 0$$

9. Find constants a and b , such that the function $f(x) = \begin{cases} \frac{x^2-4}{x-2}, & \text{if } x < 2 \\ ax^2 - bx + 3, & \text{if } 2 \leq x \leq 3 \\ 2x - a + b, & \text{if } x \geq 3 \end{cases}$

is continuous for all values of x .

To be cont. $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$ for all x , points of discontinuity 2, 3 since everywhere else $f(x)$ is continuous in its domain

$$\lim_{x \rightarrow 2^-} \frac{(x+2)(x-2)}{(x-2)} = 4 \quad \text{has to be equal}$$

$$\lim_{x \rightarrow 2^+} ax^2 - bx + 3 = 4a - 2b + 3$$

$$4a - 2b + 3 = 4$$

Equation

$$4a - 2b = 1$$

$$8a - 4b = 2$$

$$9a - 3b + 3 = 6 - a + b$$

$$10a - 4b = 3$$

$$\rightarrow -(8a - 4b) = -2$$

$$2a = 1 \quad \text{plug back in}$$

$$a = \frac{1}{2}$$

$$a = \frac{1}{2} \quad b = \frac{1}{2}$$

$$8(\frac{1}{2}) - 4b = 2$$

$$4 - 4b = 2$$

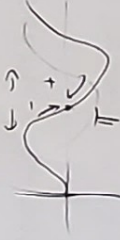
$$b = \frac{2-4}{-4}$$

$$b = \frac{-2}{-4}$$

(T: 4 marks)

%

$$a = \frac{1}{2} \quad b = \frac{1}{2}$$



$$\lim_{x \rightarrow 1} \frac{5 \sin(\pi x)}{2(1-x^3) \left(\frac{1}{5 \sin(\frac{\pi}{x})} \right)}$$

$$\lim_{x \rightarrow 1} \frac{5 \sin(\pi x)}{2(1-x)(x^2+x+1)}$$

$$\text{as } x \rightarrow 1^+ \quad \frac{5 \sin(\pi x)}{2(1-x)(x^2+x+1)} \quad \text{goes to } +\infty$$

$$\text{as } x \rightarrow 1^- \quad \frac{5 \sin(\pi x)}{2(1-x)(x^2+x+1)} \quad \text{goes to } -\infty$$

$$\lim_{x \rightarrow 1} \frac{5 \sin(\pi x)}{2-2x^3} = \infty$$

NOTE: Please write all the solutions in order. Show all your work for full marks.

The graphs of functions f and g are shown below, on the interval $-3 \leq x \leq 5$. The graphs consist of line segments, except on the interval $0 < x < 2$.

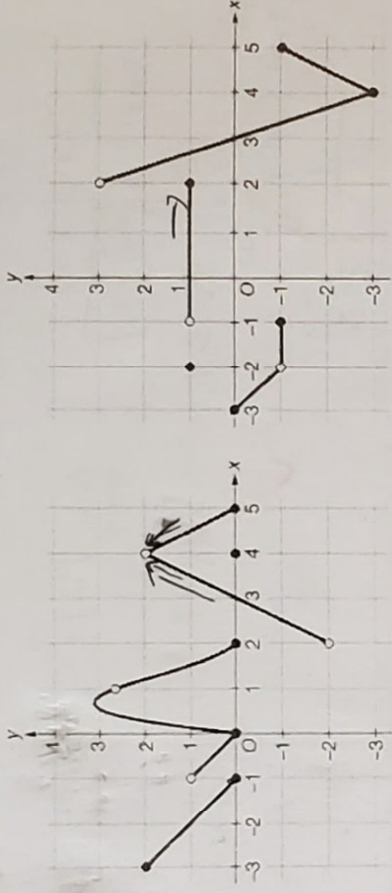
(A: 6 marks)
(2)

What types of discontinuity points do functions f and g have and at which points?

Function f :

Jump at $x = -1$ and $x = 2$

Hole/Removable at $x = 1$ and $x = 4$



Graph of f

Function g :

Jump at $x = -1$ and $x = 2$

Hole/Removable at $x = -2$

Graph of g

Let $h(x) = g(f(x))$. Using properties of limits, determine $\lim_{x \rightarrow 4} h(x)$. Use correct limit notation in your answer.

$$\lim_{x \rightarrow 4} g(f(x))$$

$$= \lim_{x \rightarrow 4} f(x)$$

$$= \lim_{x \rightarrow 4} f(x) = 2$$

$\lim_{v \rightarrow 2^-} g(v) = 2$ $g(v)$ approaches 2 as v approaches 2 from the left

Let $k(x) = f(x) + g(x)$. Consider $x = -1$ and $x = 2$. At which one of these values does $k(x)$ have a jump discontinuity? Justify your answer for the jump discontinuity using correct limit notation. (2)

For $x \rightarrow -1^-$

$$\lim_{x \rightarrow -1^-} k(x)$$

$$= \lim_{x \rightarrow -1^-} f(x) + \lim_{x \rightarrow -1^-} g(x)$$

$$= \lim_{x \rightarrow -1^-} f(x) + \lim_{x \rightarrow -1^-} g(x)$$

$$= -1 + 2 = 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$= 1$$

$$\lim_{x \rightarrow 2} f(x) = -1, \lim_{x \rightarrow 4^+} f(x) = -\infty, \lim_{x \rightarrow 4^-} f(x) = \infty$$

$$\lim_{x \rightarrow \infty} f(x) = \infty, \lim_{x \rightarrow -\infty} f(x) = 2.$$

